



Question Set

Question 1

Consider a down-and-out call monitored discretely and continuously. Define the state variables (spot path, barrier level, first passage time, and terminal payoff) and derive the first-passage and knock-out valuation condition. State the measure, filtration, and tradability assumptions.

$$\tau_H = \inf \{t \geq 0 : S_t \leq H\}$$

Question 2

For a down-and-out call monitored discretely and continuously, explain which probability measure is natural for pricing. Write the density-process logic needed if the starting model is under the physical measure.

Question 3

Identify the discounted tradable or density-adjusted process in a down-and-out call monitored discretely and continuously. Prove the martingale condition and name one modelling choice that would break it.

Question 4

Using spot 100, strike 100, barrier 80, volatility 22 percent, and one-year maturity, construct a two-step numerical calculation. Show intermediate state values before discounting or aggregating.

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Question 5

For a down-and-out call monitored discretely and continuously, compute and interpret barrier delta and gamma instability near the barrier. Explain the hedge sign and the unit of each sensitivity.

Question 6

Suppose the desk observes vanilla surface inputs needed before applying barrier corrections. Write the calibration objective, the weights you would use, and the parameter-identifiability checks.

Question 7

State the terminal and boundary conditions for a down-and-out call monitored discretely and continuously. Explain how absorbing barrier and terminal call payoff prevents economically impossible prices.

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Question 8

Design a Monte Carlo or lattice estimator for a down-and-out call monitored discretely and continuously. Specify Brownian-bridge correction between monitoring dates, the estimator, the discounting rule, and the standard error.

Question 9

A junior quant implements a down-and-out call monitored discretely and continuously and gets a stable but wrong answer. Explain how underestimating knock-out probability with coarse time steps creates the error and design a diagnostic test.

Question 10

Write an NumPy algorithm for a down-and-out call monitored discretely and continuously. Include inputs, outputs, diagnostics, and one benchmark case.

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Question 11

Starting from the payoff definition of a down-and-out call monitored discretely and continuously, derive a replication, dynamic-programming, or conditional-expectation representation without skipping the economic argument.

Question 12

Create a stressed-market version of spot 100, strike 100, barrier 80, volatility 22 percent, and one-year maturity. Recompute the value or risk metric and explain which term dominates the change.

Question 13

Use a second-order Taylor approximation for a down-and-out call monitored discretely and continuously. Separate linear exposure, convexity exposure, cross terms if relevant, and residual model risk.

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Question 14

Show how a change of numeraire or change of measure would simplify one expectation in a down-and-out call monitored discretely and continuously. Identify what becomes a martingale after the change.

Question 15

Give a proof-style answer showing when the wealth process for a down-and-out call monitored discretely and continuously is self-financing. Explain why predictable trading strategies matter.

Question 16

Market quotes for vanilla surface inputs needed before applying barrier corrections are noisy. Propose a robust calibration workflow with initial guesses, constraints, residual plots, and out-of-sample checks.

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Question 17

For a down-and-out call monitored discretely and continuously, compare the mathematical boundary condition with the trader's intuition. Give one limiting case where the answer should be obvious.

Question 18

Design a convergence study for Brownian-bridge correction between monitoring dates. Specify path counts, time-step refinement, confidence intervals, and the exact quantity plotted.

Question 19

Find one way that underestimating knock-out probability with coarse time steps could pass a superficial unit test. Write a stronger test that would fail for the biased implementation.

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Question 20

Explain a down-and-out call monitored discretely and continuously and code the core routine. Give the derivation outline, pseudocode, and three numerical sanity checks.